



Effect of attrition on particle size distribution and SO₂ capture in fluidized bed combustion under high CO₂ partial pressure conditions

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ABSTRACT

In both pressurized and oxygen-enriched fluidized bed combustion the partial pressure of CO₂ in the reactor becomes high, which affects SO₂ capture by limestone. Both of these technologies are also applicable to decreasing greenhouse gas emissions; the first one by increasing the efficiency of electric energy production and the latter by enabling capture of carbon dioxide for storage.

Attrition increases the reaction rate by removing the sulphated layer on the particle, thus reducing the diffusion resistance. In the well-known solution for the shrinking core model the reaction time can be presented as the sum of the contributions of the kinetics and diffusion. It is shown that the effect of attrition can be expressed as an auxiliary term in this expression. A method to extract the diffusivity of the product layer from the SO₂ response in a bench-scale fluidized bed test using a limestone sample with a wide particle size distribution is presented. Based on a population balance model, a method to estimate the particle-size-dependent attrition rate from measured particle size distributions of the feed and bed material is illustrated for a 71-MW_e pressurized power plant. In addition attrition and its effect on the optimization of the limestone particle size for sulphur capture in oxygen-enriched combustion are discussed.

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1. Introduction

Pressurized fluidized bed combustion (PFBC) is an established technology. On the other hand the oxygen-enriched FBC with flue gas recirculation is becoming topical, since it creates a high concentration of CO₂ in the flue gases, making it easier to separate CO₂ for sequestration. Buhre et al. (2005) have reviewed the technology using oxygen-enriched gas instead of air. The production of oxygen has an energy penalty. In a recent study the overall net efficiency in power production drops from 43.4% to 33.3–34.9% (Myöhänen et al., 2009). Most of the earlier studies on combustion in O₂/CO₂ atmospheres and its effect on NO_x and SO_x emissions concern pulverized coal. In addition, a number of studies have been reported on laboratory-scale circulating fluidized bed combustion (CFBC) experiments with high concentrations of oxygen (Czakiert et al., 2005; Myöhänen et al., 2009; Fang et al., 2007; Jia et al., 2007; Pikkarainen et al., 2007), some of them also include sulphur emissions. Recycling of flue gases (mainly CO₂ and H₂O) applied in O₂ combustion affects NO and SO_x levels in the reactor and their emissions. In the literature, SO_x

emissions are usually reported to be lower at high CO₂ partial pressure conditions. The opposite, however, has also been reported. CaO particles are much lighter than CaCO₃ particles, which affects the location and loss of the particles in the boiler. The primary fragmentation, which is much more intense in an O₂/N₂ atmosphere, may lead to a significant loss of non-reacted fine particles.

Limestone is widely used in FBC not only as an SO₂ absorbent but also as a bed material to transport heat from the fuel particles to boiler tubes. In both of these FBC technologies, the partial pressure of CO₂ in the reactor becomes high, which affects SO₂ capture by limestone. At high of CO₂ partial pressure conditions the sulphur capture takes place by the direct sulphation (DS) reaction (SO₂ + CaCO₃ + $\frac{1}{2}$ O₂ = CaSO₄ + CO₂) without the calcination step (CaCO₃ = CaO + CO₂). DS has been reviewed (Hu et al., 2006). Pseudo-steady-state shrinking (unreacted) core models (SCM) (see e.g. Szekely et al., 1976) have been shown to describe reasonable well the DS of single limestone particles in laboratory-scale devices. SCM has been widely employed to describe the DS of single limestone particles.

In FBC, attrition removes the sulphated layer thus increasing the rate of sulphur capture. There is some evidence that attrition may be more pronounced in DS (Shimizu et al., 2001). Liu et al. (2000), in analysing the sulphated product layer in DS found it to be rather fragile. Scala and Salatino (2008) found that in DS the attrition rate

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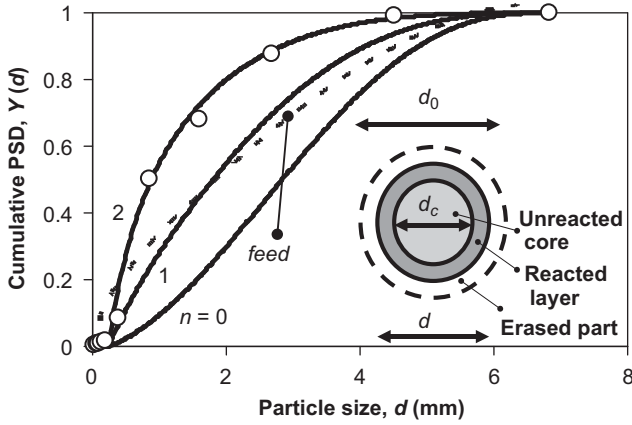


Fig. 1. Cumulative PSD, $Y(d)$, in the bed with different values of n . Dots (O) are measured points. Inside figure: A reacting particle with decreasing particle radius due to attrition.

increases with time in contrast to the strongly decreasing behaviour in an O_2/N_2 atmosphere.

Particle size distribution (PSD) is an important factor in FBC and in CFBC, affecting the fluidization behaviour, heat transfer and pollutant formation. The effect of attrition on PSD and sulphur capture in FBC in atmospheres with high CO_2 partial pressure conditions is studied. Though the effect of reactions on PSD in FBC (Kunii and Levenspiel, 1991) has been studied using population balance modelling (PBM), attrition rather than reaction is the main mechanism affecting PSD in the case of limestone particles. The diameter is reduced by attrition at the rate $R(d) = \partial d / \partial t$ (Fig. 1). This is generally correlated with a power dependence on the diameter by $R(d) = -Kd^n$, where K is a constant (depends on excess fluidization velocity) and n is an experimental exponent. This exponent strongly affects the calculated PSD (see Fig. 1). The particle diameter may also change due to reactions, since the molar volume of the product is larger than that of the initial material. Here it is assumed that the product layer is strong, preventing an increase in the diameter by reactions and the change is only due to attrition. The change in molecular volume with reaction is not so large for DS. There seems no systematic research on attrition using a wide range of particle sizes although different values for the exponent n (0...2.6) have been reported in the literature for limestone (Franceschi et al., 1980). According to them one reason for the different values reported for the exponent n is due to experimental conditions and the fact that the minimum fluidization velocity also depends on the particle size ($\sim d^{0.5...2}$ or Reynolds number). Attrition tests for a relatively wide range of particle sizes indicate the value $n = 1$ (Kono, 1981), since the fractional mass loss rate was practically independent of particle size.

In this study, we apply a bench-scale FBC reactor to study the chemical reactivity of limestone (Tsukumi, cited in Shimizu et al., 2001) with SO_2 in a feed gas mixture of CO_2 and O_2 (and SO_2). In addition, we developed an SCM accounting for attrition to describe the sulphur capture. To calculate the average conversion of a particle during its lifetime in a CFBC boiler, we incorporate a particle model with a residence time distribution (RTD) model. The PSDs of the limestone in the bed are calculated using PBM and attrition is studied based on the PSDs measured. To analyse the results reported for a 71-MW_e pressurized FBC (Shimizu et al., 2001), we then apply PBM. We also extend the optimization of particle size in VTT's 50-kW pilot-scale circulating FBC reactor, studied previously by Saastamoinen (2007), to conditions of oxygen-enriched combustion with high partial pressure of CO_2 .

2. Model description

2.1. Population balance modelling for particle size distribution

The evolution of the PSD of limestone is described by the PBM for limestone particles undergoing attrition

$$\frac{\partial P(d, t)}{\partial t} = \dot{m}_0 p_0(d, t) - k_{\text{loss}}(d)P(d, t) - \frac{\partial [R(d)P(d, t)]}{\partial d} + 3P(d, t)R(d)/d \quad (1)$$

Here $P = mp$, m is the mass of Ca (or alternately the volume of limestone could be used) in the bed, p its normalized PSD and $\dot{m}_0$ is the feed mass flow rate of Ca.

The term on the left-hand side of Eq. (1) describes the change of the PSD in the FBC boiler, which is zero in steady state. The first term on the right-hand side describes the particle feed into the boiler, the second is the loss in fly ash and bottom ash drain, the third accounts for the reduction of particle size by attrition and the last is due to fine attrition particles, which are assumed to escape immediately after their generation from the furnace, if there is no cyclone or the cut size of the cyclone is large as in the case studied here. A more complete equation accounting for the effects of PSD of fines and the cyclone has been presented (Saastamoinen et al., 2006).

2.2. Shrinking core model for reaction with simultaneous attrition

We consider a particle undergoing an SCM-type reaction with simultaneous removal of material from the surface by attrition (Fig. 1). The rate of conversion can be derived from the pseudo-steady-state balance for the limestone particle, by stating that diffusion through the shell surrounding the unreacted core is equal to the chemical reaction rate at the core and that the reaction leads to shrinking of the core. The rate of conversion can be presented as

$$\frac{dX_*}{dt} = \left(\frac{\tau_c}{3} (1 - X_*)^{-2/3} + 2\tau_D [(1 - X_*)^{-1/3} - 1/f(t)] + \tau_M/f(t) \right)^{-1} \quad (2)$$

when $X_* < 1$ with conditions $X_*(t=0) = 0$ and $(dX_*/dt)_{X_* \geq 1} = 0$. The time constants τ_c , τ_D and τ_M correspond to the complete conversion time with the control of chemical kinetics, diffusion and surface mass transfer without the effect of attrition, respectively. All the material may not be accessible for reactions, which is taken into account by the maximum conversion X_{max} . Eq. (2) differs from the generally known form in that it contains the function $f(t) = d/d_0$, describing the reduction of diameter by attrition as a function of time. It can be iteratively integrated by presenting it in a form that shows the required time to reach a given conversion. By assuming $\tau_M \ll \tau_D$, we get

$$t = \tau_c [1 - (1 - X_*)^{1/3}] + \tau_D [1 + 2(1 - X_*) - 3(1 - X_*)^{2/3}] - 2\tau_D \int_0^{X_*} [1/f(t) - 1] dX \quad (3)$$

On the right-hand side of this equation the first and second well-known terms are the contributions of chemical kinetics and diffusion, respectively. The last additional term is the reduction of the required time due to attrition. The conversion is defined as that for the whole material that was originally in the particle (inside the dashed line in Fig. 1), $X_* = 1 - (d_c/d_0)^3$. If attrition fines are assumed to have reached the maximum conversion X_{max} , the degree of conversion of the remaining material in the particle that has escaped in bottom ash flow is $X_{*b} = 1 - (d_c/d)^3 = 1 - (1 - X_*)^{1/3}/[f(t)]^3$. If diffusion is limiting the process, the time constant for kinetics can be approximated as $\tau_c = 0$. Further, if the diffusion rate across the surface layer is slow

($\tau_D \rightarrow 0$), the reaction becomes completely controlled by attrition. In order to have a product layer, the second term in Eq. (2) must be positive giving the condition $f(t) > (1 - X_*)^{1/3}$. In the limiting case of attrition control, this condition gives $X_* = 1 - [f(t)]^3$, which leads to the values $X_{*A} = 1$ and $X_{*b} = 0$. For the chemical control we get $t/\tau_C = 1 - (1 - X_*)^{1/3}$, i.e. $X_* = 1 - (1 - t/\tau_C)^3$, which corresponds to the reaction diameter $d_c = d_0(1 - t/\tau_C)$. Then in order to guarantee that the surface is fully converted before it is removed by attrition we should have $d_c < d$, i.e. $f(t) = d/d_0 \geq d_c/d_0 = 1 - t/\tau_C$. However, due to attrition the chemical reaction front also proceeds faster. In the case of a not fully converted surface, the conversion of attrition fines is the same as that of the particle surface, $X_A = X_{\max} G^{-1}(\Pi^2)$, where G^{-1} is an inverse function defined elsewhere (Saastamoinen and Shimizu, 2007). For constant values (not depending on local conversion X inside a particle) of k_0 , n and D_s , $G(X) = X^2/2$. The conversion at the surface becomes in this special case $X_A = \sqrt{2}X_{\max}\Pi$. The condition for a fully converted product layer is $\Pi > \sqrt{G(1)}$. The steady-state mass of the sorbent and inert material (ash, sand) in the bed is

$$\frac{m_{\text{sorbent}}}{\dot{m}_{0,\text{Ca}}} = \int_0^\infty \left(\frac{M_{\text{CaCO}_3}}{M_{\text{Ca}}} (1 - X_b) + \frac{M_{\text{CaSO}_4}}{M_{\text{Ca}}} X_b \right) f^3(t) g(t) dt,$$

$$\frac{m_{\text{inert}}}{\dot{m}_{0,\text{inert}}} = \int_0^\infty f^3(t) g(t) dt \quad (4)$$

The first term in the brackets is the contribution of CaCO_3 and the second is that of CaSO_4 . Note the misprint (last X_b missing) in earlier equations (Saastamoinen, 2007). Eventually limestone escapes from the reactor in the bottom and fly ash. The SCM gives conversion X while the RTD model accounting for attrition (Saastamoinen, 2007) is applied. The average conversion of a size class is found by combining the SCM and RTD

$$\bar{X} = \int_0^\infty [E_b(t)X_b(t) + E_A(t)X_A(t) + E_{fm}(t)X_b(t)] dt \quad (5)$$

where $E_b(t) = f^3(t)g(t)/\tau_b$, $E_A(t) = -3f^2(t)f'(t)g(t)$ and $E_{fm}(t) = f^3(t)g(t)/\tau_C$. The effect of PSD is found by summing up (or integrating) the contributions of different size classes.

3. Discussion

3.1. Determination of effective product layer diffusivity from SO_2 response in a bench-scale fluidized bed with a wide PSD

The PSD of Tsukumi limestone used in the experiments is shown in Fig. 2a. The bench-scale FBC (height of the tube 1.52 m, inner diameter 32.8 mm) is described elsewhere (Pikkarainen et al., 2007). The measured SO_2 responses in the experiments with a small batch of limestone are shown in Fig. 2b. The conversion based on SO_2 responses was calculated by

$$\frac{dX}{dt} = \frac{M_{\text{CaCO}_3} \Delta \dot{m}_{\text{SO}_2}}{M_{\text{SO}_2} m_{\text{CaCO}_3}} \quad (6)$$

where $\Delta \dot{m}_{\text{SO}_2}$ is the mass consumption rate of SO_2 in the reactor. The bench-scale fluidized bed experiments were also examined by using Eq. (2), which can be integrated numerically for each particle size. The description of reactivity in DS by an SCM makes the analysis of experiments easier, since the SCM gives directly the particle size dependency of reactivity in contrast to the reactivity descriptions generally applied for sulphation in an O_2/N_2 atmosphere, where a narrow particle size range must be applied. The logarithmic average SO_2 concentration in the bed is obtained by using measured

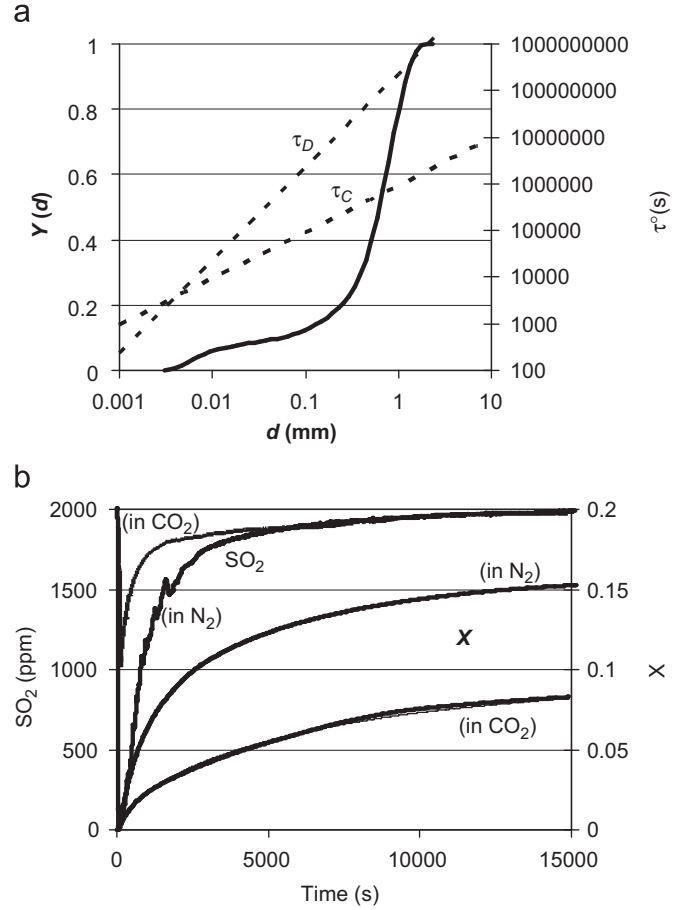


Fig. 2. (a) PSD of Tsukumi limestone used in the experiments (solid line). Time constants τ_C and τ_D (calculated with $D_s = 9.7 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$ and $k_c = 0.0003 \text{ ms}^{-1}$ in O_2/CO_2 atmosphere with 2000 ppm SO_2). (b) Measured responses of SO_2 and calculated conversions in a bench-scale fluidized bed reactor with a limestone sample $m_0 = 20.34 \text{ g}$ with gas feed 12.42 l min^{-1} (NTP) in 94 vol% CO_2 , and 6 vol% O_2 at temperature 1123 K. For comparison the case with N_2 instead of CO_2 is also shown. The inlet SO_2 concentration is 2000 ppm. Conversions calculated by Eq. (6) are shown by thick lines, conversion (in CO_2) calculated by Eq. (7) is shown by a thin line (which is so close to the thick one that it is difficult to see it separately).

inlet (in) and outlet (out) concentrations of SO_2 , $\bar{\rho}_{\text{SO}_2}(t) = (\rho_{\text{SO}_2,\text{in}} - \rho_{\text{SO}_2,\text{out}}) / \ln(\rho_{\text{SO}_2,\text{in}} / \rho_{\text{SO}_2,\text{out}})$. The conversion for the whole PSD will be

$$\bar{X} = \int_0^{d_{\max}} X[t, d_0, \bar{\rho}_{\text{SO}_2}(t)] p_0(d_0) dd_0 \quad (7)$$

The values for diffusivity D_s and chemical reactivity k_c are found by seeking values so that Eqs. (6) and (7) give the best agreement. It is assumed that the density of the limestone $\rho_{s,0} = 2700 \text{ kg m}^{-3}$ and $X_{\max} = 1$. D_s is the major parameter affecting the shape of the conversion curve. However, k_c also affects it during the initial stage, when the fine particles are reacting more rapidly. In the later stage, fine particles are totally converted. The values $D_s = 9.7 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$ and $k_c = 0.0003 \text{ s}^{-1}$ (reaction order $n_c = 1$) gave a good agreement. The D_s value is close to that reported by Hajaligol et al. (1988) at a high concentration of CO_2 . The k_c value is lower (about 0.001 s^{-1} at 1123 K) than that of Hajaligol et al. (1988). The k_c value obtained here is not very accurate due to the wide PSD used in the experiments.

The SO_2 concentration decreases close to zero in the initial stage in the N_2 case and close to half of the initial level in the CO_2 case indicating a much lower conversion rate at a high partial pressure of CO_2 (Fig. 2). The reason for the higher rate in an O_2/N_2 atmosphere is the higher diffusivity of calcined limestone. The rate in an O_2/CO_2

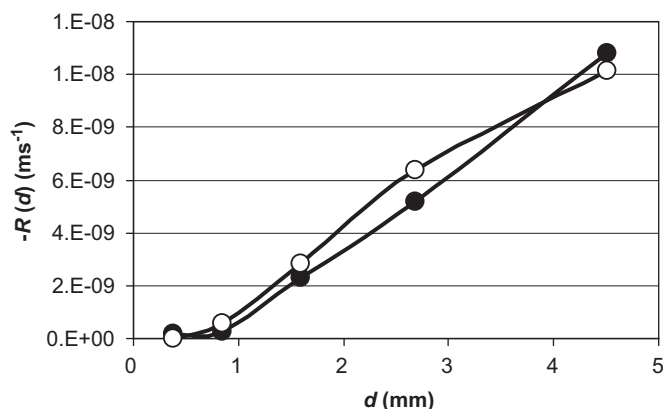


Fig. 3. Calculated rates of reduction of particle diameter by attrition based on measured PSDs of the feed and in the bed for cases studied by Shimizu et al. (2001). Phase 1 (●, average of 11 tests) and Phase 2 (○, average of four tests).

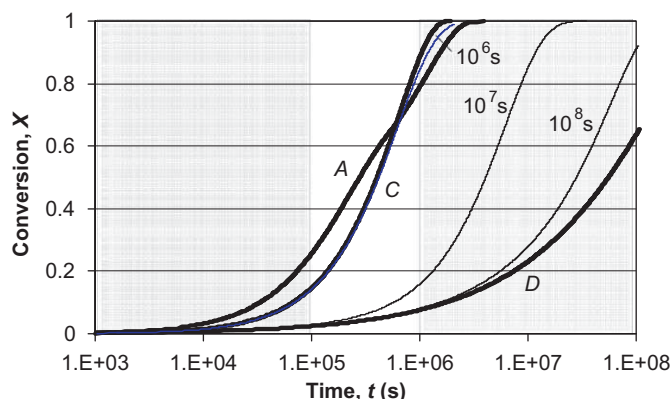


Fig. 4. Calculated conversion of limestone particle as function of time: A attrition control with attrition parameters for Phase 1; C, chemical control; D diffusion control (combined control by diffusion and kinetics practically coincides with diffusion control), combined control with attrition (thin lines), when $R_0 = 0 \text{ m s}^{-1}$ and $\tau_A = 10^6$, 10^7 and 10^8 s . $d_0 = 1 \text{ mm}$, $\rho_{\text{SO}_2} = 0.00148 \text{ kg m}^{-3}$, $\tau_c = 1.95 \times 10^6 \text{ s}$, $\tau_D = 5.02 \times 10^8 \text{ s}$. The average residence time range 10^5 – 10^6 s (Shimizu et al., 2001) in the 71-MW_e pressurized boiler is shown by the non-grey area.

atmosphere (based on SO_2 responses) exceeds that in N_2 only after 5700 s. The reason for the higher reactivity in N_2 for such a long time length and the higher conversion in the end of the experiment is probably fragmentation in the initial stage. The primary fragmentation in an O_2/CO_2 atmosphere is minor whereas in O_2/N_2 it can be substantial due to calcination (Saastamoinen et al., 2008).

3.2. Attrition and sulphur capture in a large scale pressurized power plant

A calculation example for steady-state PSD by solving Eq. (1) (using the numerical method of Saastamoinen et al., 2006) is shown in Fig. 4. The measured values of the PSDs of feed and bed particles are from 71-MW_e PFBC (Shimizu et al., 2001). The calculations were continued until a steady state was reached. The feed and the total attrition rates of Ca and the total mass of Ca in the bed were known and the values of K ($= 2.625 \times 10^{-8} \text{ m s}^{-1}$ for $n = 0$, $4.345 \times 10^{-6} \text{ s}^{-1}$ for $n = 1$, $1.064 \times 10^{-3} \text{ m}^{-1} \text{ s}^{-1}$ for $n = 2$) were chosen so that the total given attrition rate was reached. The value $n = 2$ seems to give the best fit in this case.

Instead of assuming a priori a power dependence and seeking a value of n giving the best agreement with the measured PSD in the

bed, it is possible to derive from Eq. (1) a direct dependence of the attrition rate $R(d)$ on measured PSDs of the feed, $p_0(d)$, and in the bed, $p(d)$, in steady state. $R(d)$ can be calculated by using the equation

$$R(d) = -(\dot{m}_{0e}/m) \frac{d^3}{p(d)} \int_d^\infty \frac{p_{0e}(x)}{x^3} dx \quad (8)$$

which is of the same form as the earlier equation (Saastamoinen et al., 2003) for the size reduction of burning char particles, but here the size reduction takes place by attrition instead of combustion. $p_{0e}(d)$ is the net feed of limestone into the boiler combined from the two first terms in the right-hand side of Eq. (1) in a stationary situation so that $\dot{m}_{0e} = \dot{m}_0 - \dot{m}_d$ and $p_{0e}(d) = [\dot{m}_0 p_0(d) - \dot{m}_d p_d(d)] / \dot{m}_{0e}$. The integral was calculated numerically using the measured PSDs. Calculation results assuming negligible fragmentation by thermal shock are shown in Fig. 3. The attrition rate depends on the conditions and there were significant differences between different cases. This new method to determine the attrition rate gives a rather linear increase in the attrition rate (Fig. 3) when the particle diameter is above a critical value. The two results indicating $n = 1$ (Fig. 3) above a critical size and the previous $n = 2$ (Fig. 1) for the whole range are not in disagreement, because the approximation $n = 2$ applies better for the whole range. The attrition rate is comparable with the average (0.2 – 0.7 nm s^{-1}) reported by Shimizu et al. (2001), since the major volume and consequently the major surface area is under 1 mm size (see Fig. 1). The attrition rate can approximately be presented as $R(d) = R_{cr}$, when $d \leq d_{cr}$, and $R(d) = R_{cr} - K(d - d_{cr})$, when $d > d_{cr}$. Then $f(t) = a + (1 - a) \exp(-t/\tau_A)$, where $a = (d_{cr} + \tau_A R_{cr})/d_0$ and $\tau_A = 1/K$, when $d > d_{cr}$. When $d \leq d_{cr}$, $f(t) = d_{cr}/d_0 - R_{cr}(t - t_{cr})/d_0$, where $t_{cr} = -\tau_A \ln[d_{cr}/d_0 - a]/(1 - a)$. Approximate values $d_{cr} = 0.71 \text{ mm}$, $R_{cr} = -0.24 \times 10^{-9} \text{ m s}^{-1}$ and $K = 2.9 \times 10^{-6} \text{ s}^{-1}$ ($\tau_A = 3.45 \times 10^5 \text{ s} = 95.8 \text{ h}$) are obtained for Tsukumi limestone in this specific boiler in Phase 1 tests.

It is assumed that the effect of attrition in bench-scale reactor tests (Fig. 2) was minor due to the relatively mild operation conditions and the small particle size so that the diffusivity and chemical reaction values obtained from these tests are not affected by attrition. The effects of diffusion, chemical kinetics and attrition are shown in Fig. 4. The particles should reside an unrealistic long time in the bed to reach a reasonable degree of conversion, if diffusion is controlling. The combined effect of diffusion, kinetics and attrition assuming $f(t) = \exp(-t/\tau_A)$ is also shown. It can be seen that without attrition, the degree of conversion during the residence time in the boiler and consequently the efficiency of sulphur capture would become very low.

3.3. Effect of attrition on optimum limestone size under high partial pressure of CO_2

The optimization (in CO_2/O_2) is similar to that presented earlier for an N_2/O_2 atmosphere (Saastamoinen, 2007), though significant differences exist. The description of VTT's pilot-scale CFBC reactor can be found elsewhere (Pikkarainen et al., 2007; Myöhänen et al., 2009). The optimization of the PSD of the limestone feed in PFBC is discussed elsewhere (Shimizu and Saastamoinen, 2007). The following short study is only a theoretical one, since the characteristics (diffusivity, chemical reactivity and attrition rate) of the limestone so far tested in this pilot-scale reactor are not known. The optimum size of limestone particles to reach a given sulphur-capture efficiency of η_S with minimum use of limestone (minimum value of β) is sought. Alternative optimization problem is to find the particle size giving the maximum efficiency for sulphur capture with a given amount of limestone (see Saastamoinen, 2007). Another optimization problem would be to seek the best limestone with a given PSD from a set of different types of limestone (of known characteristics) for minimizing the amount or cost of limestone. To simplify the prob-

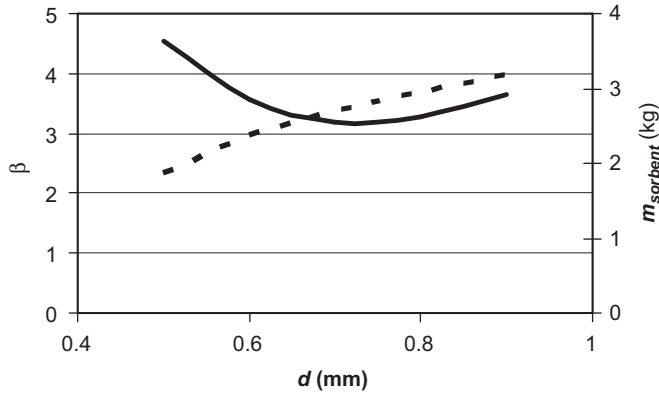


Fig. 5. Dependences of β (solid line), τ_b (dashed line) and mass of sorbent (dotted line) in the bed on particle size, when bed mass is constant, $m_b = 5$ kg. Particle size decreases due to attrition $dd/dt = -d/\tau_A$, where $\tau_A = 100$. In calculations $D = D_0 + (D_1 - D_0)X$, $D_0 = 1 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$, $D_1 = 1 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$, $k_i = 5.509 \text{ kg}^{0.84} \text{ m}^{-2.52} \text{ s}^{-1}$, $n_i = 0.16$, $X_{\max} = 1$, $\rho_{s,0} = 2700 \text{ kg m}^{-3}$, $\dot{m}_S = 2 \text{ g s}^{-1} = 0.014 \text{ g s}^{-1}$ (in fuel), $\dot{m}_{\text{inert}} = 0.1 \text{ g s}^{-1}$, $d_{\text{inert}} = 0.6 \text{ mm}$, $\rho_{\text{SO}_2} = 0.002 \text{ kg m}^{-3}$ and $\eta_S = 0.7$.

lem it is also assumed that particles of different size experience the same average SO_2 concentration. This assumption is not necessarily exactly valid, since large particles can reside for the greater part of their residence time in the bed area and small ones in the riser and in the return loop. However, the optimum particle size range is rather narrow and consequently the differences in this narrow range are small. The cyclone time constant $\tau_c = \tau_i / (1 - \eta_c)$ depends on the cyclone efficiency $\eta_c = 1 / [1 + (0.0859/d)^{2.6}]$ and particle residence time in the loop $\tau_p = 8.0 - 12.7d + 615d^2$ (units are s and mm) (Saastamoinen, 2007). It characterizes the residence time of a particle in the boiler if the particles exit only via cyclone (i.e. zero bottom ash drain).

SCMs have widely been shown to predict well the laboratory-scale results in DS at high partial pressure of CO_2 (see e.g. Hajaligol et al., 1988; Hu et al., 2006). An SCM was used to analyse the experiment from the bench-scale reactor to determine the product layer diffusivity. However, the application of this product layer diffusivity and SMC in pressurized conditions (Fig. 4) shows that the SCM without considering attrition is not necessarily well suited to modelling sulphur capture in large scale fluidized beds. The erosive conditions in CFBC are even harder than in bubbling FBC. The concept of product layer diffusion cannot explain the relatively high efficiencies for sulphur capture reached with VTT's pilot-scale CFBC reactor (Pikkarainen et al., 2007; Myöhänen et al., 2009).

In the procedure for finding a minimum value of β for a given mass flow rate of S to the boiler, it is required that the bed mass, from Eq. (4), is $m_b = m_{\text{sorbent}} + m_{\text{inert}} = \text{constant}$. Shimizu et al. (2001) showed that the reaction is confined to a very thin layer in DS under attrition. Then the bottom ash conversion can be approximated by $X_b \approx 1 - [1 - 2(\delta_c + \delta)/d]^3$. δ is the effective thickness of the layer under reaction and δ_c is the thickness of the fully reacted layer (Saastamoinen and Shimizu, 2007). $\delta = X_{\delta} \sqrt{D/k_i}$, when $n_i = 1$ and diffusivity and chemical reaction rate coefficient are constant. Here $X_{\delta} = X_{\max} \sqrt{2\Pi}$, when $\Pi < 1/\sqrt{2}$ and $X_{\delta} = X_{\max}$ for fully converted product layer. For the more complex case dealt with in Fig. 5, the approximation $\delta \approx 2 \times 10^{-12} - R^{-0.724}$ can be found, if fully converted product layer is formed (in units m and s). $G(X) = -X/b - \ln[1 - bX]/b^2$ giving $X_A/X_{\max} \approx \Pi/(0.734 + 0.396\Pi^{1.4})$, when $b = 1 - D_1/D_0 = 0.9$. The condition for fully converted product layer is $\Pi > 1.32$. The mass flow rate of Ca required to reach a given efficiency can be calculated by $\beta = \eta_S/\bar{X}$, where $\bar{X}$ is defined by Eq. (5). Eq. (4) gives a relation between particle initial diameter d_0 and the required time constant τ_b for bottom ash removal. The integrals can be calculated

numerically. The optimum particle size d_0 is found as the one giving the minimum value for β (Fig. 5).

4. Conclusions

PBM is applied to calculate the PSD formed under attrition in FBC. The effect of attrition on PSD and sulphur capture in a commercial PFBC is also discussed. The dependence of attrition rate on particle size is determined from using measured PSDs of the feed and in the bed by an explicit expression that is derived from PBM. For particles over a critical size the attrition rate is found to increase directly proportional to the particle diameter.

The SCM is applicable to analyse DS in laboratory-scale devices. However, it seems not to be adequate for modelling of sulphur retention by limestone particles in FBC under DS conditions. Application of the SCM without taking attrition into account underestimates the sulphur capture and the degree of utilization of limestone. The diffusivity of the product layer consisting of essentially the same compound (CaSO_4) is two orders of magnitude lower in DS compared with the one generated from calcined limestone particles (Hajaligol et al., 1988). However, even if the diffusivity is higher in DS, it is not high enough to explain large-scale experiments. The relatively high extent of limestone utilization obtained (Myöhänen et al., 2009; Pikkarainen et al., 2007) in an O_2/CO_2 atmosphere can be explained by strong attrition removing the product layer. An alternative explanation would be that the diffusivity becomes higher due to cracks formed in a more erosive environment.

The optimization of the particle size of limestone in an O_2/CO_2 atmosphere is discussed. Equations for determining the optimum particle size are presented. There is a need for measurements of the product compounds in the bed. The present theory predicts that the major compound in the bed originating from the limestone feed is CaCO_3 and that CaSO_4 is only present on narrow surface regions of the particles. Attrition is treated as an independent process in the paper. A dense layer with a sharp face is formed near the surface in the reaction, which may cause stress in non-porous particles inducing attrition.

Notation

D	diffusivity, $\text{m}^2 \text{ s}^{-1}$
d	particle diameter, m
$E(t)$	residence time distribution, s^{-1}
$f(t)$	function describing change of particle diameter with time under attrition; $f(t) = d/d_0$
$g(t)$	function for particle number in the bed, $g(t) = \exp[-\int_0^t ((1/\tau_b) + (1/\tau_c)) dt]$
k	reaction rate coefficient, m s^{-1} for k_c , $\text{kg}^{1-n} \text{ m}^{-3(1-n)} \text{ s}^{-1}$ for k_0
k_{loss}	coefficient for particle loss by fly and bottom ash flow, s^{-1}
K	attrition coefficient, $\text{m}^{1-n} \text{ s}^{-1}$
m	mass, kg
$\dot{m}$	mass flow rate, kg s^{-1}
M	molar weight, kg kmol^{-1}
n	attrition exponent, reaction order
$p(d)$	normalized particle distribution (based on mass of Ca), $p(d) = dY(d)/dt$, m^{-1}
$P(d)$	particle size distribution, $P(d) = m_{\text{Ca}} p(d)$, kg m^{-1}
$R(d)$	rate of change of diameter, $R(d) = d_0 f'(t)$, m s^{-1}
Sh	Sherwood number
t	time, s
T	temperature, K

X	degree of conversion ($X_* = X/X_{\max}$, $\bar{X}$ average, $X_{\max}$ maximum)
$Y(d)$	cumulative weight fraction

Greek letters

β	molar ratio $[Ca]/[S]$
η	efficiency, for sulphur capture $\eta_S = \beta \bar{X} = \bar{X}(\dot{m}_{Ca}/M_{Ca})/(\dot{m}_S/M_S)$
Π	dimensionless parameter, $\Pi = -2(M_{CaCO_3}/M_{SO_2})\sqrt{D_s k_{0i} \rho_{SO_2}^{(n_i+1)/2}}/[\sqrt{n_i+1} \rho_{s,0} R(d)]$
ρ	mass/volume, concentration, density, kg m^{-3}
τ	time constant, τ_A for attrition, $\tau_b = m_b/\dot{m}_b$ for bottom ash removal, τ_c for cyclone; complete conversion time, $\tau_D = (X_{\max}/24)(M_{SO_2}/M_{CaCO_3})(\rho_{s,0}/\rho_{SO_2})d_0^2/D_s$ for diffusion control, $\tau_C = (X_{\max}/2)(M_{SO_2}/M_{CaCO_3})(\rho_{s,0}/\rho_{SO_2})d_0/(k_c \rho_{c,SO_2}^{n_c-1})$ for control of kinetics, $\tau_M = (X_{\max}/6)(M_{SO_2}/M_{CaCO_3})(\rho_{s,0}/\rho_{SO_2})d_0^2/(ShD_g)$ for control of mass transfer, s

Indexes

O	inlet, initial
A	attrition
b	mother, bottom ash
c	reaction, reaction front, cyclone
cr	critical
fm	mother particles in fly ash
g	gas, convection
i	intrinsic
s	solid

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